

CENG381 Stochastic Processes

Martingales — Practice 1

Q1 (30 points). A stochastic process $\{Z_n; n \geq 1\}$ is a martingale if (i) $E[Z_n] < \infty$ for all $n \geq 1$, and (ii) $E[Z_n | Z_{n-1}, \dots, Z_1] = Z_{n-1}$ for all $n \geq 2$.

- a) Let $X_1, X_2, \dots$ be i.i.d. random variables with $E[X_i] = 0$. Define $S_n = X_1 + X_2 + \dots + X_n$ ($S_0 = 0$). Prove that $\{S_n; n \geq 0\}$ is a martingale by verifying condition (ii). Which property of X_i is essential, and why?
 - b) For the symmetric random walk, $X_i = +1$ w.p. $1/2$ and -1 w.p. $1/2$. Write $S_n = S_{n-1} + X_n$. Compute $E[S_n | S_{n-1}]$ directly and confirm the martingale condition. What is $E[S_n]$ for all $n \geq 0$?
 - c) Now suppose $X_i = +1$ w.p. p and -1 w.p. $q = 1 - p$, with $p \neq 1/2$. (i) Compute $E[X_i]$ and $E[S_n | S_{n-1}, \dots, S_1]$. (ii) Is $\{S_n\}$ a martingale when $p > 1/2$? Call it a sub-martingale if $E[Z_n | \text{past}] \geq Z_{n-1}$. Classify $\{S_n\}$ for $p > 1/2$ and for $p < 1/2$.
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Q2 (35 points). Let $\{S_n; n \geq 0\}$ be the symmetric random walk with $S_0 = 0$ (steps ± 1 w.p. $1/2$ each). Define $Z_n = S_n^2 - n$.

- a) Prove $\{Z_n; n \geq 0\}$ is a martingale. Compute $E[Z_n | S_{n-1}]$ by expanding $S_n = S_{n-1} + X_n$, using $E[X_n] = 0$ and $E[X_n^2] = 1$.
 - b) Let $T = \min\{n \geq 1 : |S_n| = N\}$ for a fixed integer $N > 0$ (first exit time from the interval $(-N, N)$). The Optional Stopping Theorem (OST) states that for a bounded stopping time, $E[Z_T] = Z_0$. Apply OST to $\{Z_n\}$: substitute $Z_T = S_T^2 - T$ and $Z_0 = 0$ to find $E[T]$.
 - c) By symmetry, $P(S_T = +N) = P(S_T = -N) = 1/2$. Confirm this using OST applied to $\{S_n\}$ (also a martingale): $E[S_T] = S_0 = 0$, and S_T takes values $+N$ and $-N$. Compute $E[T]$ for $N = 4$. What are the units?
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Q3 (35 points). A gambler starts with $\$s$ ($0 < s < N$). Each round they win $\$1$ w.p. $1/2$ or lose $\$1$ w.p. $1/2$ (fair game). The game stops at time T when their fortune first reaches $\$0$ (ruin) or $\$N$ (goal).

- a) The fortune process $\{S_n\}$ is a martingale. Apply OST ($E[S_T] = S_0 = s$) to find the probability $p_s = P(\text{reach } N | \text{start at } s)$. Hint: $S_T = N$ w.p. p_s and $S_T = 0$ w.p. $1 - p_s$.
- b) Apply OST to the quadratic martingale $\{Z_n = S_n^2 - n\}$: $E[Z_T] = Z_0 = s^2$. Express $E[S_T^2]$

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in terms of p_s and N , substitute p_s from (a), and derive $E[T] = s(N - s)$.

c) A gambler starts with \$3 aiming for \$7 ($N = 7$). Find the probability of reaching the goal, and the expected number of rounds until the game ends.